

```

NUMPY
data visualization:
data visualization is used to show the data

```

```

pip install numpy

```

```

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)

```

```

import numpy as np
print(np.__version__)

```

```

2.0.2

```

```

a=np.ndarray([1,2,3,4])
print(a)

```

```

[[[-0.25 -0.5   0.   -0.25]
 [ 0.25 -0.5   0.5  -0.25]
 [ 0.25  0.    0.5   0.25]]

 [[ 0.25  0.5   0.    0.25]
 [-0.25  0.5  -0.5   0.25]
 [-0.25  0.   -0.5  -0.25]]]

```

```

a=np.array([1,2,3,4])
print(a)

```

```

[1 2 3 4]

```

```

b=np.array([[1,2,3],
            [4,5,6],
            [7,8,9]])
print(b)

```

```

[[1 2 3]
 [4 5 6]
 [7 8 9]]

```

```

#3d tensor
c=np.array([
    [
        [1,2],
        [4,3],
    ],
    [
        [2,1],
        [4,3],
    ]
])
print(c)

```

```

[[[1 2]
  [4 3]]

 [[2 1]
  [4 3]]]

```

```

#built in array creation fun
#zeros and ones
d=np.zeros([3,3])
print(d)

```

```

[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]

```

```

d=np.ones([3,3])
print(d)

```

```

[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]

```

```
d=np.full([3,3],7)
print(d)
```

```
[[7 7 7]
 [7 7 7]
 [7 7 7]]
```

```
e=np.eye(4)
print(e)
```

```
[[1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]
```

```
f=np.arange(2,10,4)
print(f)
```

```
[2 6]
```

```
np.linspace(3,10,5)
```

```
array([ 3. ,  4.75,  6.5 ,  8.25, 10.  ])
```

```
#random arrays
np.random.rand(3,3)
```

```
array([[0.97233508, 0.01330845, 0.99236844],
       [0.43863939, 0.78439067, 0.34766284],
       [0.90026345, 0.37207297, 0.14941536]])
```

```
np.random.randint(3,60,size=(3,4))
```

```
array([[44, 22, 25, 47],
       [11,  5, 25, 59],
       [23, 35,  4,  4]])
```

```
np.empty((4,3))
```

```
array([[5.10503145e-315, 0.00000000e+000, 2.10077583e-312],
       [6.79038654e-313, 2.22809558e-312, 2.14321575e-312],
       [2.35541533e-312, 6.79038654e-313, 2.22809558e-312],
       [2.14321575e-312, 2.46151512e-312, 2.41907520e-312]])
```

```
arr=np.array([["a","b","c"],[1,2,3]])
print(arr.shape)
print(arr.ndim)
print(arr.size)
print(arr.dtype)
print(arr.itemsize)
print(arr.nbytes)
```

```
(2, 3)
2
6
<U21
84
504
```

```
a=np.array([1,23,7,3,4,5,51])
print(a[0])
print(a[-1])
print(a[-3])
```

```
1
51
4
```

```
print(a[1:4])
print(a[:5])
print(a[::4])
print(a[::-2])
print(a[::-1])
```

```
[23  7  3]
[ 1 23  7  3  4]
[1 4]
[51  4  7  1]
[51  5  4  3  7 23  1]
```

```
m=np.array([[1,2],
            [3,4]])
#single elements:[row,col]
print(m[0,1])
print(m[1,0])
print(m[1,1])
```

```
2
3
4
```

```
#row slicing
print(m[0,:])
print(m[1,:])
```

```
[1 2]
[3 4]
```

```
#col slicing
print(m[:,0])
print(m[:,1])
```

```
[1 3]
[2 4]
```

```
a=np.array([10,2,0,30,1,7,24])
#boolean mask
mask=a>10
print(mask)
print(a[mask])
```

```
[False False False  True False False  True]
[30 24]
```

```
# shorthand
print(a[a>15])
#fancy indexing
print(a[[1,3,5]])
```

```
[30 24]
[ 2 30  7]
```